TECHNOLOGY ETHICS • COMPLETE STUDY NOTES

Set 3: Moral Considerations, Ethical Dilemma and Conflict

Law vs Ethics, Resolving Ethical Dilemmas, 7-Step Decision Template, Simon & Aggie Cases, Group Dilemma Cases

PART 3

Law, dilemmas and case analysis

3 sections, about 2 h 30 min of reading.

Law versus ethics, and resolving ethical dilemmas

~40 min to read | High priority | 11 questions

Source: Set 4, slides 2-10; Set 3, slides 4-12; Set 2, slides 41-43

The nature of ethics

Set 4 opens with its aims. The lecture sets out to teach the fundamental principles and canons of engineering ethics, to clarify the nature of ethics, to give a process for resolving ethical dilemmas, to cover the professional bodies' model rules of conduct and the code of ethics, and to use case studies so that you examine problems from a second-person perspective (you advise or judge someone else's choice). The first two of those aims are what this section covers. For the model rules and codes see t09 and t11.

The slide on the nature of ethics makes two statements.

- Ethics is generally concerned with **rules or guidelines for morals and/or socially approved conduct**.
- Ethical standards generally apply to **conduct that can or does have a substantial effect on people's lives**.

The second point matters for engineers. A badly chosen bolt or a hidden software flaw touches many lives, so engineering conduct falls squarely inside the field of ethics. Notice that the first statement says "morals and/or socially approved conduct". Ethics is not only private conscience (morals); it also covers what a group or society approves of. That is why professional codes count as ethical standards.

Fuller definitions of ethics, morals, values and law, and the theories, are in t01.

Law versus ethics

The slides (taken from a CSUN senior ethics lecture) set law and ethics side by side. Both guide conduct, but they work differently.

Law	Ethics
Creates rules to guide conduct	Offers guidance on conduct
Balances competing values	Addresses situations in which competing values clash
Punishes conduct that is "illegal" through formal structures (courts, police, penalties)	Incentives and disincentives may be created by a "group", formal or informal (a professional body, an employer, peers, society)

Read the table row by row and you have the answer to "what is the difference between ethics and law".

- **Source and form.** Law is a set of rules made by an authority (the state). Ethics is guidance, not a command backed by the state.
- **Handling of values.** Law tries to balance competing values and settle them into a rule. Ethics deals with the moment when those values clash and no rule has settled the matter.
- **Enforcement.** Law punishes through formal structures: courts, fines, imprisonment. Ethics is enforced by groups, either formally (a professional body can discipline a member) or informally (loss of reputation, social disapproval). The rewards and penalties are incentives and disincentives, not sentences.

beyond the slides Other differences students usually add: law is written, public and applies to everyone in a territory, while ethics may be unwritten and differs between people and cultures; law often sets a minimum standard, ethics a higher one; and law can change by amendment while ethical convictions change slowly.

> **Key point.** Law punishes illegal conduct through formal structures. Ethics offers guidance where competing values clash, and its incentives and disincentives come from a group, formal or informal.

> **Q:** QB E3 · 7 marks **What is the difference between ethics and law?**

Is "legal" the same as "ethical"?

The slides give both sides of the argument.

YES, legal is ethical	NO, legal is not the same as ethical
Law defines duties, rights and "allowable conduct".	Law does not address all ethical dilemmas.
This is the **compliance approach** to business ethics: fulfil legally recognised duties, and do not go further.	Legal duties may not meet the standard of ethical conduct.
	This is the **"beyond compliance" approach**: fulfil legally recognised duties, but do not stop there.

The slide ends with the conclusion you should memorise: **in case of conflicts, it is generally held that legal standards must give way to ethical standards.**

A company on the compliance approach asks "Is it allowed?". A company on the beyond-compliance approach asks "Is it right?". An engineer who follows only the minimum the law demands can still harm people, for instance by meeting a legal emission limit while knowing a cheap fix would cut the harm further.

> **Watch out.** Do not write that law and ethics are the same, or that they are unrelated. They overlap (murder is both illegal and unethical) but neither contains the other. Say that law does not cover everything and that ethics wins in a conflict.

Law versus morality: the four categories

Set 2 warns "don't confuse the two". It first asks a clicker question: which of the following ensure that behaviour is ethical? I. Following the law. II. Acting in the best interest of society. III. Following non-legal standards for socially appropriate conduct. The options are all of the above, II and III only, none of the above, or I only. The answer is **II and III only** beyond the slides (the slide prints the question only; the answer follows from the categories on the next slides). Following the law alone does not guarantee ethical conduct, but acting in society's best interest and following non-legal standards of socially appropriate conduct do.

Then it crosses legal/illegal with moral/immoral to get four categories and one example each.

Category	Example from the slides
Legal and moral	Designing a system to be safe.
Legal and immoral	Owning a slave before the American Civil War (slavery was lawful then).
Illegal and moral	Parking in a no-parking zone to come to the aid of an injured person.
Illegal and immoral	Killing an innocent person.

The two off-diagonal cells prove law and morality are different things. A law can be immoral (slavery), and an illegal act can be the moral one (breaking a parking rule to save a life). When you need an example "where the two differ", use one of those two.

Q: Set 2 clicker **Which of the following ensure that behaviour is ethical? I. Following the law. II. Acting in the best interest of society. III. Following non-legal standards for socially appropriate conduct.**

Q: 2017 Q3D · 6 marks (also 2021-22 exam Q2B) **Explain the difference between moral and legal responsibility (mention two points of difference). Illustrate your explanation with an example where the two differ.**

MARKER WORDS: ETHICS VERSUS LAW

Say it like this:

> Law **creates rules to guide conduct**, **balances competing values** and **punishes illegal conduct through formal structures**. Ethics **offers guidance on conduct** and **addresses situations in which competing values clash**, with **incentives and disincentives created by a group**. Where they conflict, **legal standards must give way to ethical standards**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
formal structures	saying how law is enforced	**SHOULD**
competing values	describing what law balances and what ethics faces in a dilemma	**SHOULD**
penalties / sanctions	contrasting legal with moral responsibility	**MUST**
compliance approach and **beyond compliance**	explaining why "legal" is not always "ethical"	**SHOULD**
does not address all ethical dilemmas	giving a limit of law	**SHOULD**
legal standards must give way to ethical standards	concluding on a conflict between the two	**SHOULD**
legal but immoral / illegal but moral	giving an example where the two differ	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

The three features of law (Set 4, Law vs Ethics).
Creates rules to guide conduct; balances competing values; punishes conduct that is "illegal" through formal structures.

The three features of ethics (Set 4, Law vs Ethics).
Offers guidance on conduct; addresses situations in which competing values clash; incentives and disincentives may be created by a "group" (formal or informal).

The compliance approach to business ethics.
Fulfil legally recognised duties, and don't go further.

The "beyond compliance" approach.
Fulfil legally recognised duties, but don't stop there.

The four law-versus-morality categories with their examples.
Legal and moral: designing a system to be safe. Legal and immoral: owning a slave before the American Civil War. Illegal and moral: parking in a no-parking zone to help an injured person. Illegal and immoral: killing an innocent person.

Consider this: "You and Al"

This is the lecture's running example. You are the manager of Big-Mart, a large discount retailer. You recently fired Al, a sales clerk, after Al punched a customer during a dispute in the store. Al admitted it after the customer complained. A manager at a competitor, Mega-Mart, phones you. Al has applied for a job there, and the manager asks whether Al is "good with customers". The slide asks one thing: what do you do?

The later slide "Legal vs. Ethical: You and Al" turns this into a grid. Across the top are Legal and Illegal; down the side are Ethical and Unethical. Reading the slide:

	Legal	**Illegal**
Ethical?	Tell the truth: "Al admitted to punching a customer."	Contact another store, X-Mart, to warn about Al.
Unethical?	"No comment."	Say "He is great with customers."

The slide gives only these four example replies. Its row labels carry question marks ("Ethical?", "Unethical?"), so it invites class discussion rather than stating verdicts.

beyond the slides A likely reading of the grid: telling the truth about a fact Al himself admitted is lawful and honest. Warning a store that did not ask goes beyond the request, and a negative reference without proper basis can be open to a legal claim, even though the motive (protecting customers) is good. "No comment" is lawful because nothing false is said, but it may ignore a possible harm to Mega-Mart's customers. Saying "He is great with customers" is dishonest and could create liability for misrepresentation.

The point of the example is that there are several defensible answers, each protecting some value (honesty, fairness to Al, safety of customers, protection of your own firm) at the expense of another. That clash is what makes it a dilemma. Compare your own reply with the eight steps and the evaluation theories below.

beyond the slides The slides do not say how a reference should be handled. A reasonable approach is to stay with facts, state only what is documented, follow the company's reference policy and, if asked directly about conduct, avoid opinion while not hiding the incident.

Q: `Set 4 slide 6 · case` **You are the manager at Big-Mart. You fired Al, a sales clerk, after he punched a customer. A Mega-Mart manager phones and asks if Al is "good with customers". What do you do? Use the steps for resolving a dilemma.**

What is an ethical dilemma?

Set 4 asks "An ethical dilemma?" and lists its features.

- A **choice has to be made**.
- It implicates **competing values, rights and goals**.
- There is **potential harm to the decision maker**.
- There is **potential harm to others**.
- There is a **"ripple effect"**: the decision has long-term, far-reaching implications.

Set 3 gives the classic definition: **a dilemma is a problem offering two possibilities, neither of which is practically acceptable.** Combine the two and you have a good answer: an ethical dilemma is a situation in which a choice must be made between options that implicate competing values, rights and goals, where each option carries potential harm to the decision maker or to others and may have a ripple effect.

Example in engineering terms: a manager tells you to approve a design that meets the budget but is not fully safe. Approving breaks your duty to the public; refusing risks your job and your colleagues' jobs. Neither is practically acceptable, and either choice has consequences beyond today. The slides' bridge and nuclear-reactor cases, covered in t07, are examples of the same thing.

Q: `CAT 2021-22 Q2d (also 2020-21 paper)` **Briefly describe how ethical dilemmas arise in the practice of the Engineering profession.**

The Pentium case and "going beyond the code"

Set 3 reviews the Pentium case from the previous lecture and asks three questions. The slides do not retell the case, so here it is.

`beyond the slides` In 1994 a mathematician found that Intel's Pentium processor sometimes gave slightly wrong answers in floating-point division. Intel had found the flaw itself and judged it so rare that it would affect almost nobody, and at first it said it would replace chips only for users who could prove they needed accuracy. Customers, the press and large buyers such as IBM reacted angrily, and the company's reputation suffered. Intel then offered to replace the chip for anyone who asked, at a cost of hundreds of millions of dollars.

The three discussion questions and a way to answer each:

1. **What could Intel have done to satisfy customers and minimise negative publicity?** Disclose the flaw promptly and honestly, say how rare it is and who is affected, and offer replacement or a fix up front rather than after public pressure.

2. **A warning such as "This product may contain unexpected flaws and might not operate correctly under all conditions". Does this solve the ethical problems for the company?** The slide's clicker offers Yes, Sort of or No and says there is no right answer. The strongest reasoning is that it does not: a blanket warning is not honest disclosure of a known flaw, it shifts risk to the buyer, and it does not remove the duty to be truthful and protect users' interests.
3. **Was there really an ethical dilemma? If so, what was it?** Yes: Intel's duty to be honest and to treat customers fairly against the cost of recalling and replacing every chip. Both options (hide the flaw, or pay for a full recall) were costly, and neither was practically acceptable. That is the definition of a dilemma on the slide.

Next the lecture moves "from codes to cases". The code of ethics for engineers gives a good set of guides to follow. But **knowing what the codes say and knowing exactly what to do in a given situation is not always obvious**. The primary reason is that really hard ethical situations **require moral reasoning and conflict resolution**. So the lecture starts with cases and practises a form of moral reasoning, which involves three things:

- taking note of which codes of engineering ethics apply;
- identifying conflicts;
- making a choice of what to do.

All of this leads to a discussion of moral considerations and moral reasoning, which becomes the case-analysis method in t07.

Q: **Set 3 slides 6-7 · discussion** **A manufacturer prints in the product literature: "This product may contain unexpected flaws and might not operate correctly under all conditions." Does this solve the ethical problems for the company?**

Q: **Set 3 slide 8 · Pentium** **Was there really an ethical dilemma in the Pentium case? If so, what was it?**

MARKER WORDS: ETHICAL DILEMMA

Say it like this:

> An ethical dilemma is a situation in which **a choice must be made** that **implicates competing values, rights and goals**, with **potential harm to the decision maker and to others** and a **ripple effect**. It is **a problem offering two possibilities, neither of which is practically acceptable**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
ethical dilemma	naming the situation; every case answer should use it	**MUST**
competing values, rights and goals	saying what clashes in the example	**SHOULD**
potential harm to the decision maker / to others	explaining why the choice is hard	**SHOULD**
ripple effect	showing the long-term, far-reaching consequences	**SHOULD**
neither of which is practically acceptable	defining a dilemma	**SHOULD**
moral reasoning and conflict resolution	explaining why codes alone are not enough	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

Define a dilemma (Set 3).
A problem offering two possibilities, neither of which is practically acceptable.

The five features of an ethical dilemma (Set 4).
A choice to be made; competing values, rights and goals; potential harm to the decision maker; potential harm to others; "ripple effect" (long-term, far-reaching implications).

The three things moral reasoning on a case involves (Set 3).
Taking note of which codes of engineering ethics apply; identifying conflicts; making a choice of what to do.

How to resolve ethical dilemmas: the eight steps

Set 4 gives eight steps, in order. Learn them as a list and be able to explain each in a line.

1. **Identify relevant facts.** Establish what actually happened and what is known, separating fact from assumption. Who, what, when, what has been agreed or promised.
2. **Identify relevant issue(s).** Name the ethical issues and the conflict, such as safety against cost or loyalty against honesty.
3. **Identify primary stakeholders.** List the people and groups affected by the decision (public, client, employer, colleagues, yourself).
4. **Identify possible solutions.** List the realistic options, including a middle path.
5. **Evaluate each possible solution.** Test every option against ethical theories and tests (next section).
6. **Compare and assess consequences.** Weigh the likely outcomes, short and long term, for each stakeholder, including harm and the ripple effect.
7. **Decide on solution.** Choose the best option and be able to justify it.
8. **Take action.** Carry the decision out. A dilemma is not resolved until you act on the choice.

The steps run in a fixed sequence: understand (steps 1 to 3), generate and test options (4 to 6), then choose and act (7 and 8). In an exam, use the step names as headings. The full 7-step analysis template used for long cases is in t07.

How to evaluate solutions: five theories and tests

Step 5 asks you to evaluate each possible solution. Set 4 offers these lenses.

Lens	The question it asks
Stakeholder / utilitarian theory	Which option brings the greatest good to the greatest number?
Rights theory	Which option respects and protects individual rights to fair and equal treatment, privacy, freedom to advance, and so on?
Justice theory	Which option gives a fair distribution of benefits and burdens? Can harm to an individual be justifiable?
Categorical imperative	"What if everyone took such action?"
"Front page test"	What if my decision was reported on the front page of the number one daily newspaper (or on social media)?

The first four are theories; the front page test is a practical check on conscience and public reputation. If you would be ashamed to see the decision in the news, that is a warning. The categorical imperative is Kant's universalisation test (see t01).

The five ethical frameworks (Set 3)

Set 3 lists a different set of five approaches, each phrased as a question. These are the lenses to use in the Set 3 style of case analysis.

Approach	Question
Rights approach	Which option best respects the rights of all who have a stake?
Utilitarian approach	Which option will produce the most good and do the least harm?
Justice approach	Which option treats people as I want to be treated?
Ethic of care approach	Which option is best for those most in need?
Virtue approach	Which option leads me to act as a responsible person?

Comparing the two lists: rights, utilitarian and justice appear in both. Set 4 adds the categorical imperative and the front page test; Set 3 adds the ethic of care (look after those most in need) and the virtue approach (act as a responsible person). Note that Set 3 words justice as "treat people as I want to be treated", while Set 4 words it as fair distribution of benefits and burdens. Quote whichever slide you are answering from, or give both.

> **Key point.** No single lens always gives the answer. Apply two or three, say where they agree, and decide. When the lenses disagree, explain which one you give priority to and why.

Q: `ethics-cat-1 Q3a · 2 marks; also asked as a 2-mark "define" in the 2011/12 and 2014/15 papers`

Explain with the aid of examples what you understand by the term ethical dilemma.

Q: `ethics-cat-1 Q3b · 6 marks` **Briefly describe the steps a given professional would take while attempting to resolve ethical dilemmas.**

Q: **CAT 2 2015 · 12 marks** Jenny recently completed her master's degree and was hired for her dream job with the local county government. In her first year she noticed that grant funds were being mismanaged and misallocated, and that coworkers used county-owned materials, including cars, for personal business. She was most shocked by the hiring practices. Applicants had to take exams proctored by government employees, and the results decided whether they were hired and for what. The proctors were letting applicants cheat because the applicants had already been chosen, many of them friends of current employees. Jenny reported this to Matt, the department's business manager and second-in-command, who said: "You heard nothing, you saw nothing; you say nothing." Jenny felt trapped. She needed the job to pay off graduate-school loans and loved the work. She also feared that leaving her first job in under a year would look bad and harm her future chances in government. But she was very uncomfortable in a culture of corruption. Describe Jenny's ethical dilemma and explain what she should do as a young professional in her circumstances.

Q: **CAT 2 2014 Q1c · 3 + 3 marks** Phillip, a trainee telecommunications engineer working for a local Internet Service Provider's Technical Department, one morning discovers that Keith, one of the company's staff, is reading e-mails to Super Dip, one of the company's clients. It is clear that this employee has been peering into hundreds of e-mails for over six months to obtain confidential details about Super Dip Ltd. Phillip has only worked 2 months at this company following his graduation. Describe Phillip's ethical dilemma. What should Phillip as a young professional do in these circumstances? [reconstructed from garbled scan]

MARKER WORDS: RESOLVING DILEMMAS AND EVALUATING SOLUTIONS

Say it like this:

> To resolve an ethical dilemma, **identify the relevant facts**, **identify the relevant issues** and **primary stakeholders**, **identify possible solutions**, **evaluate each possible solution**, **compare and assess consequences**, **decide on a solution** and **take action**. Evaluate solutions using **utilitarian**, **rights** and **justice** theories, the **categorical imperative** and the **front page test**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
stakeholders	saying who is affected by the choice	**MUST**
categorical imperative	testing an option with "what if everyone took such action?"	**MUST**
justice	asking about fair distribution of benefits and burdens	**MUST**
greatest good to the greatest number	applying stakeholder/utilitarian theory	**SHOULD**
rights to fair and equal treatment, privacy, freedom to advance	applying rights theory	**SHOULD**
front page test	checking public reputation or social media exposure	**SHOULD**
compare and assess consequences	step 6 of the process	**SHOULD**
ethic of care and **virtue approach**	using the Set 3 frameworks	**SHOULD**

▶ **Recall drill: write the eight steps from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

The eight steps to resolve an ethical dilemma (Set 4).
Identify relevant facts; identify relevant issue(s); identify primary stakeholders; identify possible solutions; evaluate each possible solution; compare and assess consequences; decide on solution; take action.

The five lenses for evaluating solutions (Set 4).
Stakeholder/utilitarian theory; rights theory; justice theory; categorical imperative; front page test.

Stakeholder/utilitarian theory (Set 4).
Greatest good to the greatest number.

Rights theory (Set 4).
Respecting and protecting individual rights to fair and equal treatment, privacy, freedom to advance, etc.

Justice theory (Set 4).
Fair distribution of benefits and burdens: can harm to an individual be justifiable?

Categorical imperative (Set 4).
"What if everyone took such action?"

The front page test.
What if my decision was reported on the front page of the number one daily newspaper (social media)?

The five ethical frameworks of Set 3 with their questions.
Rights: which option best respects the rights of all who have a stake? Utilitarian: which option will produce the most good and do the least harm? Justice: which option treats people as I want to be treated? Ethic of care: which option is best for those most in need? Virtue: which option leads me to act as a responsible person?

Case analysis and the 7-step template

~1 h 00 min to read | High priority | 9 questions

Source: Set 3, slides 13-23 (bridge case and nuclear-reactor case); the 7-step ethical analysis template (appendix); question bank A1(i), C2; 2017 paper Q4; 2015 paper Q6

> **Why this section matters.** Whenever an exam gives you a story and says "analyse", "identify the ethical issues" or "using the template as a guide", the marker is checking that you went through the seven steps in order and that you named one ethical theory and used its own vocabulary. The question bank sets two 12-mark questions of exactly this kind (Simon and Aggie). The ethical theories themselves (Kantianism, utilitarianism, social contract) are taught in section t01 and section t02. The steps for resolving a dilemma and the Pentium case are in section t06. The four group cases are in section t08.

The 7-step ethical analysis template

The template is a table with two columns. The left column is the step. The right column lists possible questions you can ask yourself at that step. You do not have to answer every question. You use them as prompts so that nothing is missed. The wording below is copied exactly from the template.

Step	What to do	Possible questions
1	Set out the facts in a fair and balanced way	Provide a short summary of the ethical scenario
2	List the main stakeholders	Who is the instigator of the problem? In this scenario did someone identify there was an ethical issue? (Not always explicit in all scenarios) Who are the stakeholders directly involved? Who are the stakeholders indirectly implicated? Who is affected by this scenario?
3	Record the major ideals and the rights that you see having a role in the decision. Select one of two ideals/rights that you see as central to the situation	Has a person's/stakeholders trust been broken? Ideals: Do the ideals of fairness/honesty/equality/duty of care have a role in the ethical decision? Rights: Does this scenario include: The right to know? The right to privacy? The right to property?
4	What obligations are there? To whom and to what is an obligation owed and what sort of obligation is it?	Laws Engineering Code of Ethics Obligations: To the profession and industry; To avoid misleading people; To produce a sound and safe product; To owners and shareholders; To foster professionalism What kind of ethical infringement has occurred for each stakeholder? Are there any laws that have been broken?
5	What are the likely consequences for each stakeholder? (major, short term, long term?)	Who benefits from the action? Who is harmed by the action?
6	Ethical theories	Social Contract Theory OR Utilitarianism OR Kantianism
7	Summary. What is the best action? What should be done?	Highlight the fact that there is often no definitive right or wrong

How to use each step in an answer

1. **Facts.** Two or three sentences. Say only what the case says. Do not argue yet. "Fair and balanced" means do not take a side while stating facts.
2. **Stakeholders.** Split them into directly involved and indirectly implicated, and name the instigator of the problem (the person or party who set the issue in motion). If a character noticed the issue (for

example Simon in the social-media case), say so.

3. **Ideals and rights.** Ideals are values such as fairness, honesty, equality and duty of care. Rights are the right to know, the right to privacy and the right to property. Say whether someone's trust has been broken. Then pick the one or two ideals or rights that are central, and say why.
4. **Obligations.** Say who owes what to whom. Use the template list: to the profession and industry, to avoid misleading people, to produce a sound and safe product, to owners and shareholders, to foster professionalism. Then state the infringement (which obligation was broken, for which stakeholder) and any law broken.
5. **Consequences.** For each stakeholder: major, short-term and long-term effects. End with who benefits and who is harmed.
6. **Ethical theory.** Choose one of social contract, utilitarianism or Kantianism. Name it, define it in one line with its marker words, and apply it to the case. Optionally add one criticism of the theory.
7. **Summary.** State the best action in one or two sentences. Then add that there is often no definitive right or wrong, so you are giving the best-justified answer.

Watch out. Question A1(i) and C2 both say "using the ethical analysis template as a guide, identify the ethical issues at play and conduct an ethical analysis using an ethical theory of your choice (12 marks)". That is two jobs: (1) the issues, which come out of steps 1 to 5, and (2) the analysis with one named theory, which is step 6, ending with step 7. Do not skip the theory. It carries the marks that students lose.

Which theory to pick, with its marker words from Set 1

Theory	Say it like this (Set 1 wording)	Apply it by asking	One criticism to add
Kantianism (Immanuel Kant, 1724-1804)	Focus on the **rightness of moral rules** (**good will**); **disregard emotional feelings and consequences**. Founded on the view that all people are **fundamentally rational beings** who can derive moral rules from the logic of the situation.	Can the rule be **universally applied to everyone**? Does the rule **treat people as ends, not means**?	It allows no exceptions to moral rules; sometimes no rule adequately characterises an action; there is no way to resolve a conflict between rules.
Utilitarianism	A **consequentialist theory**. It decides whether an act or rule is right depending on whether it results in the **increase of the aggregate happiness (utilities)**. **Act utilitarianism** judges the benefits of a single act. **Rule utilitarianism** judges the benefits of all actions that follow the rule.	Who would benefit? Who would be harmed? Do the benefits exceed the costs (or does harm outweigh benefit)?	It requires aggregating all consequences on a single scale; it does not recognise or respect individual rights (a minority could be sacrificed for the majority).
Social contract theory	Individuals **implicitly accept a social contract** on the condition that others follow the rules as well. Individuals act within a **sphere of freedom** as long as the rules are respected. The rules exist to enable everyone the **benefits of social living** and centre on respecting **natural rights to life, liberty and property** (John Locke).	Whose rights are involved? **Negative rights (liberties)** are the right to act without interference. **Positive rights (claim-rights)** are an obligation of some people to provide certain things for others. Did someone break the rules everyone implicitly accepts?	Conflicting rights problem; it may be unjust to people who are incapable of following the rules.

Rule of thumb for choosing. Use **social contract** when the case is about rights such as privacy, property or the right to know. Use **utilitarianism** when you can count benefits and harms to many people. Use **Kantianism** when the case is about a duty such as telling the truth, keeping a promise or not using people as a tool. The model answers below use all three so that you see each in action.

MARKER WORDS: THE 7-STEP TEMPLATE

Say it like this:

> The ethical analysis template has seven steps: **set out the facts** in a fair and balanced way, **list the main stakeholders**, record the major **ideals and rights**, state the **obligations**, weigh the **consequences** for each stakeholder, apply an **ethical theory** (social contract, utilitarianism or Kantianism), and give a **summary** of the best action.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
stakeholders directly involved / indirectly implicated	listing who is in the case (step 2)	**MUST**
instigator of the problem	naming who started the ethical issue	**SHOULD**
trust has been broken	opening step 3	**SHOULD**
fairness, honesty, equality, duty of care	naming the ideals	**MUST**
right to know, right to privacy, right to property	naming the rights	**MUST**
to avoid misleading people; to produce a sound and safe product	stating obligations (step 4)	**SHOULD**
ethical infringement	saying what was broken and for whom	**SHOULD**
short term, long term; who benefits, who is harmed	step 5	**MUST**
no definitive right or wrong	closing the answer (step 7)	**SHOULD**

▶ **Recall drill: write the seven steps from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

The seven steps of the ethical analysis template, in order.

1 Set out the facts in a fair and balanced way; 2 List the main stakeholders; 3 Record the major ideals and rights; 4 What obligations are there?; 5 Likely consequences for each stakeholder; 6 Ethical theories; 7 Summary: what is the best action, what should be done?

The three ethical theories offered at step 6 of the template.
Social Contract Theory, Utilitarianism, Kantianism.

The ideals named at step 3 of the template.
Fairness, honesty, equality, duty of care.

The rights named at step 3 of the template.
The right to know, the right to privacy, the right to property.

The obligations listed at step 4 of the template.
To the profession and industry; to avoid misleading people; to produce a sound and safe product; to owners and shareholders; to foster professionalism.

Kant's two criteria.
Can the rule be universally applied to everyone? Does the rule treat people as ends, not means?

Define utilitarianism.
A consequentialist theory that decides whether an act or rule is right depending on whether it results in the increase of the aggregate happiness (utilities).

Define social contract theory.
Individuals implicitly accept a social contract, on the condition that others follow the rules as well; the rules center on respecting natural rights to life, liberty and property.

Case 1: protecting the safety of society (the bridge)

The case, retold

Your employer asks you to design a bridge that must not cost more than $1 million to build. After a study you find three things.

- An ideal bridge can be built for $1.5 million.
- Given the design constraints, a bridge built for $1 million will collapse in a moderate earthquake.
- A bridge built for $1.25 million will survive a moderate earthquake but will collapse in an infrequent extreme earthquake.

Then your employer says: "If we don't build the bridge for $1 million, then we are going to have to lay off half of the staff, including you." He asks you to go ahead with the next stage of the project. What do you do?

What is the conflict?

The class was asked to pick the primary conflict from four options. The answer is the second: **your duty to society versus your loyalty to your own career**.

- The code of ethics for engineers requires you to take the **safety of society as being of paramount importance**.
- But you also feel a personal sense of loyalty to your company and fellow workers. You do not want anyone to lose their job.
- So the conflict is between your duty to society and your loyalty to your own career and the welfare of your fellow employees.

The other options were wrong. Duty to fellow employees versus duty to the boss is a side issue. Uncertainty about the largest earthquake is a technical problem, not the conflict. Honesty to clients versus finishing the project does not capture the safety danger.

What is more important?

The conflict sets your future employment and the employment of others in your company against the welfare of society. The slide gives a firm ruling: **in a case like this the welfare of society comes first**. Your duty to protect the public is greater than your duty to your own career and that of your fellow employees.

> **Key point.** The reason society wins is the engineering code: safety of the public is paramount, which means it ranks above loyalty to an employer or colleagues. Loyalty is real but secondary. State it in that order.

> **Q:** **Set 3 case 1 · template practice** **Your employer asks you to design a bridge costing no more than $1 million. A $1m bridge collapses in a moderate earthquake; a $1.25m bridge survives it but collapses in an extreme one; the ideal bridge costs $1.5m. Your employer threatens to lay off half the staff, including you, unless you proceed at $1m. Analyse the case using the ethical analysis template and one ethical theory.**

Case 2: telling the truth in public statements (the nuclear reactor)

The case, retold

The government asks you to verify that a nuclear reactor will not leak toxic substances into the neighbouring ocean. After your study you find that:

- the reactor will likely leak within the coming 8 years, but there is significant uncertainty;
- the reactor cannot be evaluated more carefully unless it is shut down immediately;
- both the ocean and the neighbouring community are at risk.

Government officials receive your report and ask you to modify it so that it says the facility is actually safe. They claim that altering the report will protect the public, preventing panic while the government tries to shut down and fix the facility. What do you do?

What is the conflict?

The code requires you to **safeguard the public's welfare**. It also requires you to **tell the truth when making public statements** concerning your area of engineering. To solve the conflict you must (1) correctly

understand what each code requirement is telling you and (2) choose to act on the obligation that is of priority.

- "Protecting the public" means making sure they are safe.
- "Issue public statements in an objective and truthful manner" means telling the public the reactor may not be safe while outlining the uncertainties.
- The government asks you to alter the report in order to protect the public. So the two duties seem to pull apart.

Your obligation as an engineer is to safeguard public safety *and* tell the truth. That means you cannot alter data and you must tell the truth about the reactor. The government is calling on you **as a citizen** to alter documents as a way of protecting fellow citizens. So the conflict is **between your obligations as an engineer and your obligations as a citizen**. This is a **role conflict**. (Role responsibilities such as engineer, employee and citizen are introduced in t04.)

What is more important?

The slide says role conflicts are hard and there is no easy answer. This is where other moral considerations matter.

- **The public's right to know.** The people near the reactor are the ones at risk. They have a right to know what the engineer found. A false "safe" report takes that right away from them.
- **The government's obligation to tell the truth.** The government owes honesty to its citizens. It cannot ask an engineer to do what it should not do itself.

The ruling: **your duty as an engineer to tell the truth when making public statements trumps your civic duty to be loyal to your government**.

> **Key point.** Both cases end the same way: a duty that comes from the engineering code (safety, truthfulness) ranks above a loyalty (to an employer, to a government). Say "role conflict" for the engineer-versus-citizen case and "welfare of society comes first" for the bridge.

> **Q:** **Set 3 case 2 · template practice** **You are asked by the government to verify that a nuclear reactor will not leak toxic substances into the ocean. You find it will likely leak within 8 years (with significant uncertainty), that it cannot be evaluated more carefully unless shut down, and that the ocean and community are at risk. Officials ask you to change your report to say it is safe, to prevent panic. Analyse the case with the ethical analysis template and one ethical theory.**

MARKER WORDS: ROLE CONFLICTS AND PRIORITY OF DUTIES

Say it like this:

> The code requires the engineer to take the **safety of society as paramount**, so in a conflict between duty to society and **loyalty** to an employer, the **welfare of society comes first**. Where the conflict is between obligations as an **engineer** and obligations as a **citizen**, the duty to **tell the truth** in public statements wins.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
paramount importance	stating what the code says about public safety	**SHOULD**
welfare of society comes first	giving the verdict in the bridge case	**SHOULD**
loyalty to your own career / employer	naming the competing duty	**MUST**
role conflict (engineer versus citizen)	explaining the nuclear case	**SHOULD**
objective and truthful manner	explaining the code on public statements	**SHOULD**
public's right to know	adding a moral consideration	**SHOULD**
trumps	saying which duty wins	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

The two competing duties in the bridge case.
Your duty to society; your loyalty to your own career and the welfare of your fellow employees.

The two competing obligations in the nuclear-reactor case.
Your obligations as an engineer; your obligations as a citizen.

Question bank A1(i): Simon and the social-media data

Q: **QB A1(i) · 12 marks** Simon is working as a Social Media Consultant for a multi-national corporation. One day he gets a call from the CIO requesting him to collect personal data on the staff in the IT department via Facebook and LinkedIn. Simon is very concerned whether it is an ethical thing to do. Using the ethical analysis template (in appendix) as a guide, identify the ethical issues at play and conduct an ethical analysis using an ethical theory of your choice.

Question bank C2: Aggie and the weed killer

Q: **QB C2 · 12 marks** Aggie's pride and joy is her garden. One day she notices weeds growing through the gaps between the pavers in her garden. She decides to buy some weed killer to kill the weeds. She selects the "Knuck-Em-Dead" weed killer spray manufactured by Knock Dead Pty Ltd. There are no instructions for application nor warnings about side effects. It turned out inhaling fumes from the weed killer causes allergic reactions in some people. The company knew this but did not put instructions or warnings for its consumers. At home Aggie sprays what she thinks is enough weed killer to kill off the weeds. That night she falls violently ill and has to be hospitalised. Doctors find she has suffered a severe allergic reaction inhaling fumes from the weed killer. She has to pay $1,000 in hospital bills. Using the ethical analysis template (in appendix) as a guide, identify the ethical issues at play and conduct an ethical analysis using an ethical theory of your choice. [wording reconstructed from garbled scan]

MARKER WORDS: APPLYING A THEORY IN A CASE ANSWER

Say it like this:

> Using **Kantianism**, the rule must be **universally applied to everyone** and must **treat people as ends, not means**. Using **utilitarianism**, the right act is the one that increases the **aggregate happiness**, so the **benefits must outweigh the harm**. Using **social contract theory**, people accept rules on condition that others do, so we must respect **natural rights to life, liberty and property**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
universally applied to everyone	testing a rule with Kantianism	**SHOULD**
ends, not means	saying a person was used as a tool	**SHOULD**
rightness of moral rules (good will)	defining Kantianism	**SHOULD**
consequentialist; aggregate happiness	defining utilitarianism	**SHOULD**
act utilitarianism / rule utilitarianism	saying whether you judge one act or a rule	**SHOULD**
social contract	defining or choosing that theory	**MUST**
natural rights: life, liberty, property	naming the rights behind social contract	**SHOULD**
conflicting rights problem	adding a criticism of social contract	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

Locke's three natural rights.
Life, liberty and property.

Negative rights (liberties).
The right to act without interference.

Positive rights (claim-rights).
An obligation of some people to provide certain things for others.

Act utilitarianism versus rule utilitarianism.
Act: judging the benefits of a single act. Rule: judging the benefits of all actions that follow the rule.

Three arguments against Kantianism.
It allows no exceptions to moral rules; sometimes no rule adequately characterises an action; there is no way to resolve a conflict between rules.

Two arguments against utilitarianism.
It requires aggregating all consequences on a single scale; it does not recognise or respect individual rights (a minority could be sacrificed for the greater happiness of the majority).

BART case (2017 paper, question 4)

The case. In March 1972 three engineers, Holger Hjortsvang, Max Blankenzee and Robert Bruder, were working on the **Bay Area Rapid Transit (BART)** project in California, United States. They were responsible for the design and creation of an automatic guided train system. They had been expressing doubts about the system's safety through **internal memos** to their managers since 1969. The managers' response was "don't make trouble". In March 1972 the three engineers were dismissed. (Part C, the four principles of the code, is in section t11. Part E, problems of law in engineering, is in section t10. Whistleblowing in detail is in section t21.)

A short template pass first, so you can see the facts that parts A, B and D use.

Q: **2017 Q4 · template walk-through** **Analyse the BART case using the seven-step template.**

Q: **2017 Q4A · 6 marks** **What would you advise Holger Hjortsvang, Max Blankenzee and Robert Bruder to do after the response by the managers? Apply two out of the following three ethical theories (deontological ethics, consequentialist ethics and virtue ethics) to decide how to act.**

Q: **2017 Q4B · 4 marks** **If the two theories come to different answers, explain why you think the outcome of one theory should prevail over the outcome of the other.**

Q: **2017 Q4D · 6 marks** **Use two different ethical theories to explain or justify the actions of a whistle-blower.**

MARKER WORDS: ADVISING THE BART ENGINEERS AND JUSTIFYING A WHISTLE-BLOWER

Say it like this:

> **Deontological ethics** judges an act by **duty and the rightness of moral rules**, not consequences, so the engineers must put **public safety first**. **Consequentialist ethics** judges by results, and the **harm to many outweighs the cost to a few**. A whistle-blower is justified because the **benefit to the public exceeds the harm to the whistle-blower**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
deontological ethics	applying the duty-based theory in 4A	**SHOULD**
universalisation	testing the managers' rule "don't make trouble"	**SHOULD**
consequentialist ethics	weighing results of speaking out or staying silent	**SHOULD**
internal memos	describing what the engineers did first	**SHOULD**
don't make trouble	quoting the managers' response	**SHOULD**
ends, not means	saying passengers must not be gambled with	**SHOULD**
whistle-blower	justifying going outside the company	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

The three theories offered in 2017 Q4A for advising the BART engineers.
Deontological ethics, consequentialist ethics and virtue ethics (apply two).

Deontological ethics (as used in the BART answer).
Judges an act by duty and the rightness of moral rules, not by the consequences.

Consequentialist ethics (as used in the BART answer).
Judges an act by whether the result gives the greatest good.

The two theories used above to justify a whistle-blower.
Utilitarianism and social contract theory.

Pacemaker and transistor case (2015 paper, question 6)

Q: **2015 Q6 · 20 marks** The heart pacemaker is a modern wonder. The device has a timer that resets itself every time the patient's heart beats. If the heart does not beat on schedule (say, within 1.2 seconds), the pacemaker gives a stimulus that causes a heartbeat. It is 1975, and you are on the board of directors of a company that makes transistors. Among the many companies with whom you have a contract is one that makes heart pacemakers. Pacemaker technology is in its infancy. When doctors implant a pacemaker, the patient's normal heartbeat is disabled, and he or she relies entirely on the device. If it fails, the patient's heart stops. Doctors are not very adept at installing the pacemakers, which are extremely delicate: there is even a story of a person yawning deeply, pulling the pacemaker wire in his chest, and dying. After that and many similar incidents, the board begins to reconsider whether your company should sell to the pacemaker company. Members of the board feel this situation is a major lawsuit just waiting to happen, and your company, as well as the company you supply, will be liable. In addition, you feel the specs the pacemaker company uses to test the transistors are not very strong. You and the board decide to get out of the business before it is too late. You tell the pacemaker company representatives about your conclusion, and they respond: "You can't stop selling us transistors. You are the sole remaining supplier for us. Everyone else has backed out for the same reasons you're giving. If you don't sell us the product, we'll go out of business. Pretty soon no one will be making heart pacemakers, and many people need them. Without pacemakers, people don't even have a chance." You take that information back to the board. People have different opinions. One says, "This is a bad deal, and it isn't our problem. We don't make enough on this sale to make the risk worthwhile." Another says, "We don't know how other companies use the transistors we sell them. Why should we be concerned about this one? What about that baby who died when the transistor in the incubator failed? We didn't know that company was using the transistor." Another says, "I think the real issue here: don't we have an ethical obligation to sell the product to the pacemaker company? What will happen if we don't sell to them?" Another says, "Give me a break. Our only obligation is to our shareholders. And how did we get so stupid that we're the last source? I'm telling you, we don't need this." Finally the chair of the board says, "OK, let's make a decision." What do you do? (a) Analyse the factual, conceptual and moral issues in the case. (b) What conclusion do you come to about the ethically correct choice that should have been made? [wording reconstructed from garbled scan]

MARKER WORDS: FACTUAL, CONCEPTUAL AND MORAL ISSUES

Say it like this:

> Factual issues are about what the facts are and whether they can be checked. Conceptual issues are about the meaning of key terms such as obligation, responsibility and risk. Moral issues are the conflicts between values, rights and duties.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
factual issues	listing the unknown or checkable facts	SHOULD
conceptual issues	asking what a word such as obligation means	SHOULD
moral issues	naming the conflict of values	SHOULD
obligation to shareholders	describing one side of the conflict	SHOULD
to produce a sound and safe product	stating the obligation to the public	SHOULD
whistle-blower / internal memos	discussing the BART engineers	MUST
harm to many outweighs cost to a few	giving the utilitarian argument	SHOULD

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

The three kinds of issue to analyse in the pacemaker question.
Factual, conceptual and moral issues.

Factual issues.
Questions about what is the case, which can be settled by evidence.

Conceptual issues.
Questions about what a key term means (obligation, responsibility, safe risk).

Moral issues.
The conflicts between values, rights and duties.

The four group cases

~50 min to read | Medium priority | 4 questions

Source: Set 4, slides 33-38 (group exercise, Cases 1 to 4, lecture recap and homework)

> **Why this section matters.** These four cases are the lecturer's own homework ("Assignments: Cases 1, 2, 3 and 4"). No past paper asks about them by name, but each one is a typical exam scenario: a short story, a question such as "what should the engineer do?", and a marker looking for the template steps, a clause from the code and one named theory. Practise each case until you can answer it in 15 minutes. The template steps are set out in full in section t07; this section uses the same seven step names. The theories are taught in section t01 and section t02. The clauses quoted come from the ASME canons and the UIPE/ERB model rules, which are taught in section t09 and section t11.

The group exercise and how to use it

Slide 33 sets the exercise. Each group must discuss one case and prepare a solution in PowerPoint. The group then has 15 minutes to present to the class and answer questions. Group 1 takes Case 1, Group 2 takes Case 2, Group 3 takes Case 3 and Group 4 takes Case 4. The slide adds one instruction: the cases are designed as problems for you, so seek resolutions that you personally can live with. That means the lecturer wants a decision you are willing to defend as your own, not a list of options.

Slide 38, the recap, repeats two ideas that every answer below relies on:

- Ethics is generally concerned with **rules or guidelines for morals and/or socially approved conduct**.
- Ethical standards generally apply to **conduct that can or does have a substantial effect on people's lives**.

The slide also sets the homework: do all four cases.

The seven template headings used for every case below

The names are the same as in the template table in t07.

1. Set out the facts in a fair and balanced way.
2. List the main stakeholders (directly involved, indirectly implicated, who is affected, the instigator of the problem).
3. Record the major ideals and rights (fairness, honesty, equality, duty of care; right to know, right to privacy, right to property).
4. What obligations are there? (laws, code of ethics, the obligations list).
5. Likely consequences for each stakeholder (short term, long term; who benefits, who is harmed).

6. Ethical theories (social contract, utilitarianism or Kantianism).
7. Summary: what is the best action, what should be done? (there is often no definitive right or wrong).

> **Watch out.** The model resolutions below are written in compact note form so you can learn the pattern. Under exam conditions you can merge steps 1 and 2 into one short paragraph if time is short, but keep the other headings visible. Each model uses one theory only, because the template says choose one. Where a second theory is mentioned it is a one-line contrast to show you understand the theory.

Case 1: safety shortcuts on presses and stamping machines

The case retold

You have just been hired as a production engineer. On the shop floor you find a possible problem. The workers are routinely ignoring some of the government-mandated safety regulations that cover the presses and stamping machines.

They do this in two ways. First, they override safety features such as guards, which are designed to make it impossible to insert a hand or arm into a machine. Second, they rig up "convenience" controls so that they can operate a machine while standing close to it, instead of using the approved safety switches. The approved switches need more movement or more operating steps.

The workers' reason, or excuse, is that following the safety features strictly would make production difficult, tiring and inefficient. They believe their shortcut is still adequately safe and gives better efficiency and worker satisfaction.

The slide asks three questions:

1. Should you immediately insist on full compliance with all the safety regulations, or do the workers have enough of a case that you would be tempted to ignore the violations?
2. If you are tempted to ignore them, how would you justify that to your boss?
3. How much weight should you give the workers' clear preference for not following the regulations? Ethically, can safety standards be relaxed if the people they apply to want them relaxed?

Model resolution using the seven template headings

Step	Model content
1. Facts	A newly hired production engineer finds workers bypassing guards and approved switches on presses and stamping machines, which breaks government safety regulations. The workers say compliance makes work hard and slow and that their shortcut is safe enough. The engineer has not yet acted.
2. Stakeholders	Directly involved: the workers (who run the risk of losing a hand or arm), the production engineer, the supervisor and the boss. The instigator of the problem is the workers' habit, which management has tolerated. Indirectly implicated: the owners and shareholders, the workers' families, the regulator (government inspectors), customers who wait for products, and insurers. Everyone who works near the machines is affected.
3. Ideals and rights	Trust has been broken: the regulator trusts the company to enforce the rules, and workers trust the company to keep them safe. Ideals: duty of care and honesty. The worker's right to life and bodily safety is central, and so is the employer's duty of care. The workers' preference for efficiency is a preference about a risk that falls on them, but it does not remove the duty of care.
4. Obligations	Laws: the safety regulations are mandatory, so the violation is already an offence. Code: ASME canon "Engineers shall hold paramount the safety, health and welfare of the public in the performance of their professional duties". UIPE/ERB model rule "While performing services, the engineer's foremost responsibility is to the public welfare". Obligations list: to produce a sound and safe product and workplace; to the profession and industry; to owners and shareholders (whose firm faces fines and claims). Ethical infringement: the company is failing in its duty of care to the workers, and the engineer would fail if he or she looked away.
5. Consequences	If you ignore the violation: short term, higher output and happier workers; long term, a likely amputation or death, compensation claims, fines, closure of the plant and loss of your licence. If you insist on compliance: short term, resentment, lower output and some cost; long term, safer workers, a legal and reputable firm. Who benefits from compliance: workers, owners, regulator, you. Who is harmed: only the short-term output and convenience.
6. Theory	Utilitarianism, a consequentialist theory that decides whether an act is right by whether it increases the aggregate happiness (utilities). Compare benefits and harms: the gain in speed and comfort is small and spread thinly, while the harm (a crushed hand or a death) is severe and irreversible, and the long-run cost to the firm is large. The harm outweighs the benefit, so compliance is right. As rule utilitarianism: the rule "obey safety regulations" produces more happiness over all cases than "obey them unless workers prefer otherwise".
7. Summary	Do not ignore the violation. First talk to the workers and supervisor, listen to the efficiency problem, then insist on the guards and approved switches. Report to your

Step	Model content
	boss and propose redesigning the workstation (for example, two-hand controls or light curtains that are safe and fast) so that safety and efficiency are both met. If management refuses, take the matter to the safety authority. There is often no definitive right or wrong, but this is the best-justified action and one I can live with.

The three questions answered directly

- **Insist or ignore?** Insist. The workers' case is about convenience, and convenience cannot outweigh risk to life.
- **How would you justify ignoring it to your boss?** You could not. A justification would have to say "the risk is small", but you cannot show that, and you would become responsible if someone was hurt. The honest thing to say to your boss is that you found a violation and propose a fix.
- **Can safety standards be relaxed because those they apply to want it?** No, for four reasons. The workers may not understand the risk fully or may be under pressure to produce (consent is not fully free). The standard also protects others: the employer, the family and the public. A regulation is a legal requirement and cannot be waived by the people it covers. And the engineer's duty is to the public welfare, which is not owned by the workers. This links to the idea of voluntary risk in section t16: a risk is only voluntary when the person knows it and freely accepts it.

If the exam asks this

Answer skeleton. (1) State the facts in two lines. (2) Name the stakeholders: workers, engineer, boss, owners, regulator. (3) Name the central right (right to life and safety) and ideal (duty of care) and say trust has been broken. (4) Quote the canon "hold paramount the safety, health and welfare of the public" and the model rule "foremost responsibility is to the public welfare"; say the regulation is law. (5) Compare short-term and long-term consequences; who benefits, who is harmed. (6) Apply utilitarianism: harm outweighs benefit. (7) Conclude: insist on compliance, redesign for efficiency, escalate if refused; add that there is often no definitive right or wrong.

Q: **Practice · 12 marks** **Workers at a plant remove machine guards to work faster, and they say they are happy to take the risk. As the new production engineer, using the ethical analysis template as a guide, analyse the case using an ethical theory of your choice and state what you should do.**

MARKER WORDS: CASE 1, SAFETY SHORTCUTS

Say it like this:

> An engineer must **hold paramount the safety, health and welfare of the public**, so workers' preference cannot relax a **government-mandated safety regulation**; the engineer owes a **duty of care** and should insist on compliance and redesign the process to be both safe and efficient.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
hold paramount the safety, health and welfare of the public	citing the canon (step 4)	**MUST**
foremost responsibility is to the public welfare	citing the model rule on obligation to society	**SHOULD**
duty of care	naming the ideal at step 3	**MUST**
trust has been broken	opening step 3	**SHOULD**
aggregate happiness (utilities)	defining utilitarianism at step 6	**SHOULD**
to produce a sound and safe product	stating an obligation	**SHOULD**
short term, long term	step 5	**MUST**

▶ **Recall drill: complete the canon and the verdict, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

ASME canon 1 (safety).
Engineers shall hold paramount the safety, health and welfare of the public in the performance of their professional duties.

Model rule on obligation to society, first clause.
While performing services, the engineer's foremost responsibility is to the public welfare.

Model rule: designs the engineer may approve.
Engineers shall approve only those designs that safeguard the life, health, welfare, and property of the public while conforming to accepted engineering standards.

Case 2: toxic liquid waste slightly over the legal limit

The case retold

You and an engineer colleague work closely on designing and carrying out procedures for the proper disposal of various waste materials in an industrial plant. He is responsible for liquid wastes, which are discharged into local rivers.

In your regular discussions you notice that he habitually allows the levels of some toxic liquid-waste chemicals to be slightly higher than the law permits for those chemicals. You tell him you have noticed. He replies that, because the levels are only slightly above the legal limit, any ethical or safety issue is trivial and not worth the trouble and expense of correcting.

The slide asks:

1. Do you agree with your colleague?
2. If not, should you try to get him to correct the excess levels, or is this none of your business since he, not you, is responsible for liquid wastes?
3. If he refuses to correct the problem, should you report it to your boss or higher management?
4. If no one in the company will do anything, should you be prepared to go over their heads and report to government inspectors or regulators directly? Or should one do that only when a much more serious risk to public health and safety is involved?

Model resolution using the seven template headings

Step	Model content
1. Facts	Two engineers share waste-disposal design in a plant. The colleague, who is responsible for liquid waste, habitually lets toxic chemical levels run slightly above the legal limit before discharge into local rivers. He says the excess is trivial and correction is not worth the cost. You have told him; he has not changed.
2. Stakeholders	Directly involved: you, the colleague, your boss and higher management. The colleague is the instigator of the problem; you are the one who identified the issue. Indirectly implicated: owners and shareholders, the environment regulator. Affected: people who drink, fish or farm with river water downstream, children, wildlife, and the plant's reputation.
3. Ideals and rights	Trust broken: the public and the regulator trust the company to respect the limit. Ideals: honesty (the plant reports compliance but is not compliant), fairness (downstream people carry the risk, the company keeps the savings) and duty of care. Rights: the community's right to know what is in its water, and its right to life and health. The company has a right to property, but not to use the river as a free dump beyond the licence.
4. Obligations	Law: the limit is set by law, so an exceedance, however small, is a breach. Code: canon "Engineers shall hold paramount the safety, health and welfare of the public"; model rule 3 on whistle blowing: "If an engineer's professional judgment is overruled resulting in danger to the life, health, welfare, or property of the public, the engineer shall notify his/her employer or client and any appropriate authority"; model rule 8, the final obligation to society: "Engineers who have knowledge of possible violation of any of the rules ... shall provide pertinent information and assist the state board in reaching final determination". Obligations list: to avoid misleading people, to produce a sound and safe product, to the profession. Ethical infringement: a breach of the duty to society; and the habit means it is not a one-off mistake.
5. Consequences	If you do nothing: short term, the company saves money; long term, toxins build up in the river and food chain (small excesses accumulate), people become ill, the company is fined or closed, and you share the blame because you knew. If you act: short term, conflict with a colleague, cost of treatment equipment, possible retaliation for you; long term, safe water, legal compliance, a firm that can defend its record. Who benefits from action: the community, the environment, the firm in the long run. Who is harmed: your relationship with the colleague and the short-term budget.
6. Theory	Kantianism, which focuses on the rightness of moral rules and disregards consequences and feelings. Universalise the colleague's rule: "It is acceptable to exceed the legal limit when the excess is small." If everyone adopted it, limits would mean nothing and every plant would drift upward; the rule cannot be universally applied to everyone and so is wrong. Also, the downstream people are treated as a means to the company's savings, not as ends. The "it is his department" rule fails the

Step	Model content
	same test, because if every engineer ignored wrongdoing by colleagues, no wrong would ever be checked.
7. Summary	Disagree with the colleague. It is your business because the harm falls on the public and the code obliges you to act. Escalate in steps: talk to the colleague again with the data and offer to help find a cheap fix; if he refuses, report to your boss and higher management in writing; if the company does nothing, report to the regulator. Do not wait for a much more serious risk: the code does not set a size threshold for a legal violation that endangers public health, although the stronger the danger, the stronger the duty. There is often no definitive right or wrong, but this is the best-justified course.

The slide's questions answered in one place

- **Do you agree that it is trivial?** No. A legal limit is a decision by society about safe levels. "Slightly over" is the colleague's opinion, and he gives no data. The excess is habitual, so it accumulates.
- **Is it your business?** Yes. The canon is about the safety, health and welfare of the public, not about whose department it is. The model rule on the final obligation to society speaks of any engineer with knowledge of a possible violation.
- **Go to the boss?** Yes, once the colleague refuses. This is the internal route and it comes first.
- **Go over their heads?** Yes if no one in the company acts. Model rule 3 names "any appropriate authority". The more serious the danger and the clearer the evidence, the more clearly you must do so. External whistle blowing is discussed fully in section t21.

If the exam asks this

Answer skeleton. (1) Facts. (2) Stakeholders: you, colleague, boss, owners, regulator, downstream community; colleague is instigator. (3) Trust broken; honesty, fairness, duty of care; right to health and right to know. (4) Law breached; canon "hold paramount the safety, health and welfare of the public"; rule 3 "notify his/her employer or client and any appropriate authority"; rule 8 final obligation. (5) Short- and long-term consequences, who benefits, who is harmed. (6) Kantianism: universalise the rule "slight excess is fine". (7) Escalate: colleague, boss, regulator; no definitive right or wrong.

Q: Practice · 8 marks **A colleague who handles effluent discharge tells you that exceeding the legal limit by a small margin is "not worth correcting". State your view and explain, with reference to the code of ethics, the steps you would take if he refuses to change.**

MARKER WORDS: CASE 2, TOXIC WASTE OVER THE LIMIT

Say it like this:

> A small excess over a legal limit is still a breach; the engineer must **hold paramount the safety, health and welfare of the public**, must **notify the employer and any appropriate authority** when danger to the public is not corrected, and under **Kantianism** the rule "slight excess is acceptable" fails because it cannot be **universally applied**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
hold paramount the safety, health and welfare of the public	citing the first canon	**MUST**
notify his/her employer or client and any appropriate authority	justifying reporting upward and outward	**SHOULD**
whistle blowing	describing going over their heads	**SHOULD**
universally applied to everyone	applying Kant's test at step 6	**SHOULD**
treat people as ends, not means	explaining why the downstream public is wronged	**SHOULD**
right to know	step 3 rights	**SHOULD**
who benefits, who is harmed	step 5	**MUST**

▶ **Recall drill: complete the model rule, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

Model rule 3 under Obligation to Society (whistle blowing).
If an engineer's professional judgment is overruled resulting in danger to the life, health, welfare, or property of the public, the engineer shall notify his/her employer or client and any appropriate authority.

Model rule 8 under Obligation to Society (final obligation).
Engineers who have knowledge of possible violation of any of the rules listed in this and the following two parts shall provide pertinent information and assist the state board in reaching final determination of the possible violation.

Model rule 7 under Obligation to Society (the "Clean Hands" rule).
Engineers shall not enter business ventures or permit their names or their firm's names to be used by any persons or firm which is engaging in dishonest, fraudulent, or illegal business practice.

Case 3: recycled-composite wall sections and whether to tell customers

The case retold

Your company has for some time supplied prefabricated wall sections, which you designed, to construction companies. One day you have a new idea: the sections could be made more cheaply from composites of recycled waste materials.

Pilot runs of the new technique are very successful, so the company decides to switch entirely to it for all future production runs. But there are management debates about how, or whether, to tell the customers about the change.

The supply contracts were written in terms of specifications and function. The load-bearing capacity and longevity of the wall sections were specified (the slide text says "love bearing", which reads as a typing error for load bearing), but no particular materials or fabrication techniques are named. So the changeover could be made without breaching any current contract.

On the other hand, the new method brings significant cost savings. The slide asks:

1. Does your company have an ethical obligation to tell the customers, and perhaps to renegotiate supply at a reduced price so that the customers also share the benefit?
2. More specifically, as a professional engineer and the designer of the new technique, do you have a special duty to be an advocate inside your company for fully informing customers about the technique and the savings?

Model resolution using the seven template headings

Step	Model content
1. Facts	You designed prefabricated wall sections. You invent a cheaper recycled-composite method; pilots succeed; management will switch all production. Contracts specify load-bearing capacity and longevity but not materials or methods, so no contract is breached by the switch. Management is unsure whether to tell customers. The savings are significant.
2. Stakeholders	Directly involved: you (inventor and designer), management, the customers (construction companies). You are the person who identified the issue. Indirectly implicated: the occupants and users of the buildings, the building inspectors and regulators, owners and shareholders, other wall-section suppliers, and the recycling and environment sector. Affected: anyone who will live or work in a building that uses the sections.
3. Ideals and rights	Honesty (disclosure of a material change), fairness (sharing the savings) and duty of care. The customers' right to know what they are buying; the company's right to property in its invention. Trust: customers trust the supplier to deliver what they expected when they signed, and silence on a change they would care about risks that trust.
4. Obligations	Law: no contract breach, but the contract is only a minimum (the letter of the contract is met; the spirit may not be, see section t10 on letter versus spirit). Code: ASME canon "Engineers shall act in professional matters for each employer or client as faithful agents or trustees"; the canon "Engineers shall issue public statements only in an objective and truthful manner"; model rule 4 "Engineers shall be objective and truthful in professional reports, statements, or testimonies and shall provide all pertinent supporting information"; the duty of full disclosure. Competence canon: "perform services only in the areas of their competence" means the new material must be tested and proved for load and longevity. Obligation list: to avoid misleading people, to produce a sound and safe product, to owners and shareholders, to foster professionalism.
5. Consequences	If you tell the customers: short term, some may ask for lower prices or reject the product, and the company earns less per unit; long term, trust and reputation, repeat business, and a track record for the recycled material. If you do not tell: short term, higher profit; long term, if customers discover it (or a defect appears) they lose trust, may sue, and the company may be blamed; the recycled technique itself may be discredited. Who benefits from disclosure: customers, end users, the company's reputation, the recycling industry. Who is harmed: short-term profit margins.
6. Theory	Social contract theory. Individuals implicitly accept a social contract on the condition that others follow the rules as well; the rules exist so that everyone has the benefits of social living. Business depends on that implicit contract: customers deal on the assumption that suppliers do not hide material changes. Hiding a cost-saving change

Step	Model content
	breaks the implicit rules of fair dealing even if it is within the written contract. The customers' rights (the right to know, the right to property in what they paid for) therefore require disclosure. A contrasting view: a utilitarian count could also favour disclosure because trust yields greater long-term benefit.
7. Summary	The company should tell the customers about the change, with test evidence that load-bearing capacity and longevity are met, and should be willing to negotiate a share of the savings. As the designer, I have a special duty to advocate this inside the company because I know the technique better than anyone and the code binds me personally as a faithful agent and a truthful reporter. If management refuses, I should state my objection in writing; I should not tell customers anything false. There is often no definitive right or wrong about price-sharing, but disclosure of a material change is the best-justified action.

The slide's two questions answered separately

- **Is there an ethical obligation to inform and perhaps to share savings?** To inform: yes, because the change is material and silence misleads by omission. To renegotiate the price: it is not strictly required by law or by the code, so it is an ethical ideal (fairness) rather than a duty. It is wise business to offer some benefit, because it builds trust. Say clearly which part is a duty and which part is good practice.
- **Special duty as designer?** Yes. Engineers have a personal professional duty that is not removed by the company's decision. See personal versus organisational responsibility in section t04. The designer is the one who can vouch that the new material meets the specification, so the designer should push for disclosure and insist on proper testing before delivery.

beyond the slides A practical point examiners like: check that building codes or standards do not name approved materials. If they do, a change of material may need regulatory approval, and that makes disclosure to the buyer and the authority a legal duty too.

If the exam asks this

Answer skeleton. (1) Facts: cheaper recycled method, contracts silent on materials. (2) Stakeholders: you, management, customers, end users, regulators. (3) Honesty, fairness, duty of care; right to know; trust. (4) Cite canon "faithful agents or trustees" and "objective and truthful"; model rule 4 "provide all pertinent supporting information"; letter versus spirit of the contract. (5) Short and long term for each party. (6) Social contract: implicit rules of fair dealing. (7) Inform customers with test evidence, offer a fair share of savings, advocate inside the company; no definitive right or wrong on price-sharing.

Q: Practice · 8 marks **Your company wants to change the material of a product without telling customers, because the contract does not name materials. Discuss whether this is ethical, using one ethical theory and one clause from a code of ethics.**

MARKER WORDS: CASE 3, RECYCLED-COMPOSITE WALL SECTIONS

Say it like this:

> A change that is within the **letter** of a contract can still breach its **spirit**; engineers must act as **faithful agents or trustees**, be **objective and truthful**, and under **social contract theory** the customers' **right to know** requires disclosure.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
faithful agents or trustees	citing the canon on employers and clients	**MUST**
objective and truthful	citing the canon or rule on statements and reports	**MUST**
provide all pertinent supporting information	citing model rule 4	**SHOULD**
the letter vs. the spirit of the law	explaining why a legal change can still be wrong	**SHOULD**
right to know	naming the right at step 3	**SHOULD**
implicitly accept a social contract	applying the theory at step 6	**SHOULD**
to avoid misleading people	stating the obligation	**SHOULD**

▶ **Recall drill: complete the canon, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

ASME canon on employers and clients.
Engineers shall act in professional matters for each employer or client as faithful agents or trustees, and shall avoid conflicts of interest.

ASME canon on public statements.
Engineers shall issue public statements only in an objective and truthful manner.

Model rule 4 under Obligation to Society (truth in duties).
Engineers shall be objective and truthful in professional reports, statements, or testimonies and shall provide all pertinent supporting information relating to such items.

The letter versus the spirit of the law.
"Read between the lines" in terms of the intent of those documents as understood by those who formulated them.

Case 4: security systems, surveillance and biometrics

The case retold

Your company makes security systems. Until now these raised few ethical problems, because the products were traditional: armed guards, locks, reinforced alloys that are hard to cut or drill, and similar methods.

As a design engineer you realise that modern technology allows much more comprehensive security packages. These could include extensive video and audio surveillance equipment and biometric monitoring devices for employees or other people seeking entry to secure areas. Biometric devices use highly personal data such as fingerprints, retinal patterns or voice patterns.

A literature search shows many ethical concerns about collecting and using such personal data. High-tech surveillance could easily become spying carried out without the knowledge of employees, and it would violate their privacy. The data gathered for security could also be sold or used outside legitimate workplace contexts by unscrupulous customers of your systems.

Your boss wants as much of this advanced technology as possible in future systems, because customers like the features and will pay well. You are concerned. The slide asks: as an engineer, do you have an ethical responsibility not to include such questionable technologies in products you design and sell, or to include them only in forms that are difficult to misuse? Or is misuse an ethical problem only for the customers who buy the equipment, and not your responsibility as an engineer?

> **Overlap with privacy.** This case is a privacy case in engineering clothing. The right to privacy, the privacy-versus-security trade-off and surveillance by governments are taught in section t18. Workplace monitoring, data mining, facial recognition and the "nothing to fear" argument are in section t20. Use their vocabulary here.

Model resolution using the seven template headings

Step	Model content
1. Facts	A security-systems maker could add video and audio surveillance and biometric entry devices (fingerprints, retina, voice) to its products. These bring risk of covert spying and of sale or misuse of the data by customers. The boss wants the features because they sell. The design engineer is concerned.
2. Stakeholders	Directly involved: you (the design engineer), your boss and the company, the customers who buy the systems (employers). The instigator of the issue is the boss's push for features, and you identified the ethical problem. Indirectly implicated: employees and visitors who are monitored, regulators and data-protection authorities, owners and shareholders, and criminals or third parties who might buy or steal the data. Affected: everyone whose body data is collected.
3. Ideals and rights	Right to privacy (central) and the right to know that one is being monitored; honesty and fairness (secret monitoring is unfair); duty of care for the data. Trust: employees trust the employer, and the customer trusts the maker, to use the data only for security. The company has a right to property in its products and customers have a right to protect their premises, so rights conflict.
4. Obligations	Law: privacy and data-protection law applies to the collection and use of personal data (see t18 and t20). Code: canon "hold paramount the safety, health and welfare of the public"; canon "perform services only in the areas of their competence"; canon "build their professional reputations on the merit of their services"; the preamble purpose "to safeguard life, health, and property, to promote the public welfare". Obligations list: to produce a sound and safe product, to avoid misleading people, to the profession and industry, to owners and shareholders. Ethical infringement: designing a product that makes spying easy, with no safeguard, would shift harm onto people who never agreed to it.
5. Consequences	If you include everything without safeguards: short term, strong sales and happy customers; long term, privacy violations, data leaks and sale of personal data, lawsuits, regulation, loss of reputation, and identity-based crimes because fingerprints and retina cannot be changed once stolen. If you include nothing: you lose sales and competitiveness, and customers may buy from a less careful maker. If you include the technology with safeguards (privacy by design): sales continue, harm is much reduced. Who benefits: the company, customers, security of premises. Who is harmed: monitored people if the data is abused.
6. Theory	Social contract theory (rights-based). Individuals implicitly accept a social contract on the condition that others follow the rules as well, and the rules centre on respecting natural rights to life, liberty and property. Employees accept monitoring at work within a sphere of freedom, and only for the stated security purpose. Covert or unlimited collection goes beyond that implicit agreement and breaks the rule of respecting the right to privacy. The maker who designs the tool shares responsibility for what it

Step	Model content
	makes easy, so it must build in limits. (Utilitarianism would ask whether security gains exceed privacy harms; its weakness is that a minority's privacy can be traded away.)
7. Summary	Do not simply refuse and do not simply sell everything. As a design engineer I do have a responsibility, because designers decide what a product makes easy. I should propose including the technologies only in forms that are difficult to misuse: visible notice of monitoring, no hidden audio, encrypted storage of biometric templates (not raw images), access logs, limited retention, no export of data and a contract clause forbidding use outside security. If management insists on a version with no safeguards, I should record my objection and decline to design the covert features. Misuse by customers is partly their ethical problem, but this does not remove the maker's duty. There is often no definitive right or wrong, but this is the best-justified course.

Why the "it is only the customers' problem" view fails

- **Foreseeability.** The case itself says misuse is easy and likely, so the harm is foreseeable. An engineer is responsible for foreseeable consequences of a design.
- **Design decides what is easy.** A product built with no logs, no encryption and a hidden-recording mode makes abuse the default. Safeguards make abuse hard.
- **Public welfare.** The canon and the preamble put the safety and welfare of the public (including monitored people) first, above the employer's preference.
- **The fair counter-point.** A product is not responsible for every use. A lock can be used by a thief to protect stolen goods. So the duty is to take reasonable steps to prevent misuse, not to guarantee that no misuse can ever occur.

beyond the slides Designers call this privacy by design or data minimisation: collect only what is needed, keep it only as long as needed, and build the limits into the product rather than leaving them to customers' goodwill.

If the exam asks this

> **Answer skeleton.** (1) Facts: surveillance and biometrics, boss wants them, risk of spying and data sale. (2) Stakeholders: you, boss, customers, monitored people, regulators. (3) Right to privacy central; honesty; trust. (4) Privacy law; canon "hold paramount the safety, health and welfare of the public"; preamble "safeguard life, health, and property, to promote the public welfare". (5) Short and long term, who benefits, who is harmed. (6) Social contract: sphere of freedom, implicit agreement, natural rights. (7) Include only in forms difficult to misuse; object in writing if refused; there is often no definitive right or wrong.

Q: **Practice · 8 marks** A design engineer is asked to add fingerprint and voice recording to a security system, knowing buyers could use it to spy on staff. Does the engineer bear any ethical responsibility for misuse by customers? Argue for one view using an ethical theory.

MARKER WORDS: CASE 4, SURVEILLANCE AND BIOMETRICS

Say it like this:

> Surveillance and **biometric** data such as fingerprints, retinal and voice patterns are **highly personal data**; the engineer must **hold paramount the safety, health and welfare of the public** and should include these technologies **only in forms which are difficult to misuse**, because covert use violates the **right to privacy**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
right to privacy	naming the central right (step 3)	**MUST**
highly personal data	describing biometrics: fingerprints, retinal or voice patterns	**SHOULD**
a form of spying, without the knowledge of employees	describing the main harm	**SHOULD**
data sold or used outside legitimate workplace contexts	describing the second harm	**SHOULD**
difficult to misuse	stating the resolution	**SHOULD**
sphere of freedom	applying social contract theory	**SHOULD**
hold paramount the safety, health and welfare of the public	citing the canon	**MUST**

▶ **Recall drill: complete the resolution, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

The three biometric examples named in Case 4.
Fingerprints, retinal patterns, voice patterns.

The purpose of the model rules (the Preamble).
To safeguard life, health, and property, to promote the public welfare, and to maintain a high standard of integrity and practice.

The two ethical concerns about surveillance data in Case 4.
Surveillance becomes spying without the employees' knowledge, violating privacy; the data is sold or used outside legitimate workplace contexts by unscrupulous customers.

The four cases side by side

Case	Core issue	Central right or ideal	Clause to cite	Theory used here	Best action in one line
1 Safety shortcuts	Workers' consent versus mandatory safety rules	Right to life; duty of care	Hold paramount safety, health and welfare of the public	Utilitarianism	Insist on compliance, redesign for efficiency, escalate if refused
2 Toxic waste over limit	Is a small legal breach trivial; is it your business	Right to health; honesty; fairness	Hold paramount safety; notify employer and any appropriate authority	Kantianism	Colleague, then boss, then regulator
3 Recycled wall sections	Duty to disclose a legal but material change	Right to know; honesty; fairness	Faithful agents or trustees; objective and truthful	Social contract	Tell customers with test evidence; offer fair share of savings
4 Security surveillance	Designer's responsibility for foreseeable misuse	Right to privacy	Hold paramount safety, health and welfare of the public; preamble	Social contract	Include only in forms difficult to misuse; refuse covert versions

The theory choices are examples. Any one of the three theories can be defended for each case if you apply its marker words correctly. See section t07 for the full list of theory vocabulary.